

2 (a)	<u>work done = force × distance</u> moved (in the direction of the force) so <u>work done in unit time = power</u> = force x distance moved in unit time (therefore power = force × velocity) {accept sensible symbols (no need define symbols), with explanation}	[1] [1]
(b)(i)	1800 = F x 12.0 F = 150 N	[1]
(b)(ii)	F = 150 N (allow ECF)	[1]
(c)	resultant force = mass x acceleration driving force – 150 = 850 × 2.50 driving force = 2125 + 150 = 2275 N	[1]
(d)	$P_{36} / P_{12} = (v_{36} / v_{12})^n$ $48600 / 1800 = (36 / 12)^n$ $27 = (3)^n$ $n = 3$	[1] [1]